Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 11)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the energy inputs and food energy outputs of three farming systems, in GJ ha^{-1} yr^{-1}.

Table 1

System	Energy input / GJ ha^{-1} yr^{-1}	Food energy output / GJ ha^{-1} yr^{-1}
Subsistence maize, hand cultivated	0.6	9.0
Intensive wheat, UK	18.0	54.0
Intensive beef, feedlot	40.0	12.0

0 1 . 1 Calculate the energy ratio (output / input) for each system in Table 1.

[3 marks]

Answer

0 1 . 2 Explain what is meant by an **energy subsidy**, and use your answers to explain why the intensive systems have much lower energy ratios.

[3 marks]

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0 1 . 3 Explain **three** ways a farmer can reduce the energy losses in a food production chain.

[3 marks]

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0 1 . 4 Explain **two** advantages of genetic modification over conventional selective breeding.

[2 marks]

0 1 . 5 Explain **four** concerns about the use of GM crops.

[4 marks]

15

0 2 . 1 Explain what is meant by **biomimetics** and describe **three** examples.

[4 marks]

0 2 . 2 Explain why many plants produce chemicals that are useful as human medicines, and give **two** examples.

[3 marks]

0 2 . 3 Explain why the discovery of medicines is a strong argument for conserving whole habitats rather than individual species.

[2 marks]

0 2 . 4 Give **one** example of a wildlife species used in physiological research and the human health issue it helps to understand.

[1 mark]

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10

0 3 . 1 A student measured the pH of a stream with a meter that reads to 0.01 pH units but had not been calibrated. Which statement is correct? Tick **one** box.

[1 mark]

A The readings are precise but may not be accurate ☐

B The readings are accurate but not precise ☐

C The readings are both accurate and precise ☐

D The readings are reliable because the meter is electronic ☐

0 3 . 2 Explain what is meant by an **anomalous result** and how a student should decide whether to include it.

[2 marks]

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0 3 . 3 Explain why a preliminary study is carried out before the main investigation.

[2 marks]

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0 4 . 1 Which of the following is an example of habitat destruction rather than a change in abiotic factors? Tick one box.

[1 mark]

A Thermal pollution of a river by a power station ☐

B Flooding a valley to create a reservoir ☐

C Lowering the water table by groundwater abstraction ☐

D Acid mine drainage entering a stream ☐

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0 4 . 2 Give **four** human activities that cause complete habitat destruction.

[4 marks]

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0 5 The Hawaiian goose (nene) was reduced by hunting, habitat loss and introduced predators to about 30 birds. A captive breeding programme started with just seven individuals and has increased the wild population to over 2000.

0 5 . 1 Explain why starting the programme with seven individuals caused problems, and describe how these problems are managed.

[4 marks]

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0 5 . 2 Explain **three** reasons, other than gene pool size, why species kept in captivity may fail to breed.

[3 marks]

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0 5 . 3 Describe how artificial insemination and embryo transfer increase breeding success in captive mammals.

[3 marks]

10

0 6 . 1 Explain **three** reasons why the development of new human activities has rarely been restricted by their environmental impacts.

[3 marks]

0 6 . 2 Explain why sedentary rural communities often exploit resources sustainably.

[2 marks]

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0 6 . 3 Explain why urban populations are often unaware that the resources they use are being depleted.

[2 marks]

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0 6 . 4 Explain why the growth of affluence has increased per capita environmental impacts.

[3 marks]

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0 7 . 1 Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

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0 7 . 2 Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

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0 7 . 3 Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

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5

0 8 A national conservation body must decide how to divide a limited budget between species.

0 8 . 1 Describe **three** criteria the IUCN uses when deciding the Red List category of a species.

[3 marks]

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0 8 . 2 A population of lemurs has fallen from 12 000 to 2 100 mature individuals. Calculate the percentage decline.

[2 marks]

Answer ..... %

0 8 . 3 Explain **three** reasons, other than its Red List category, why one species may be given a higher conservation priority than another.

[3 marks]

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0 8 . 4 Explain the difference between CITES Appendix I and Appendix II, and explain why controlling trade can protect a species more effectively than habitat protection alone.

[3 marks]

0 8 . 5 Describe **four** ways that UK legislation and designated areas protect habitats.

[4 marks]

15

0 9 . 1 Explain how each of the following environmental impacts of salmon farming arises and how it can be reduced.

[8 marks]

Overfishing of wild fish for feed

Sea lice and disease

Escapes affecting wild gene pools

Organic waste pollution

0 9 . 2 Explain why culling seals and cormorants near fish farms is controversial and give an alternative.

[2 marks]

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1 0 . 1 Describe the conservation efforts used to protect tropical rainforests and evaluate their effectiveness.

[9 marks]

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Extra space

1 0 . 2 State **one** reason why rainforest conservation is important to countries far from the tropics.

[1 mark]

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Write an essay on one of the following topics.

1 1 . 1 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

OR

1 1 . 2 Discuss how human activities influence biodiversity, using examples from a range of different contexts.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS